

Binary relations

Outline

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Binary relations

Introduction: Binary relations

- ▶ In this section we define **binary relations** as subsets of a Cartesian product
- ▶ We will see fundamental examples: “less than”, “less than or equal to”, “is a subset of”
- ▶ We will introduce the **reflexive**, **symmetric** and **transitive** properties
- ▶ Relations allow us to formalize the concepts of order and of function

Binary relation

Definition 1: of binary relation

Given two sets A and B , a **binary relation** R is a subset of the Cartesian product $A \times B$

- ▶ When we say that R is a binary relation on a single set A , we mean that R is a subset of $A \times A$.
- ▶ A binary relation will simply be called a relation (however, non-binary relations also exist).

Example

- ▶ The relation “less than” (“ $<$ ”) on the natural numbers is the set:

$$R_{<} = \left\{ (a, b) : a, b \in \mathbb{N} \text{ and } a < b \right\}.$$

- ▶ The relation “less than or equal to” (“ $\leq$ ”) on the natural numbers is the set:

$$R_{\leq} = \left\{ (a, b) : a, b \in \mathbb{N} \text{ and } a \leq b \right\}.$$

- ▶ The relation “is a subset of” $R_{\subseteq}$ on the power set of the natural numbers (denoted by $2^{\mathbb{N}}$) is the set:

$$R_{\subseteq} = \left\{ (A, B) : A, B \in 2^{\mathbb{N}} \text{ and } A \subseteq B \right\}.$$

Reflexive, symmetric and transitive properties

- ▶ A relation $R \subseteq A \times A$ is **reflexive** if:

$$\forall a \in A, \quad (a, a) \in R$$

- ▶ A relation $R \subseteq A \times A$ is **symmetric** if:

$$\forall a, b \in A, \quad (a, b) \in R \Rightarrow (b, a) \in R$$

- ▶ A relation $R \subseteq A \times A$ is **transitive** if:

$$\forall a, b, c \in A, \quad (a, b) \in R \text{ and } (b, c) \in R \Rightarrow (a, c) \in R$$

Example

- ▶ The relation $R_{\leq}$ is reflexive, but $R_{<}$ is not.
- ▶ The relations $R_{<}$ and $R_{\leq}$ are not symmetric.
- ▶ The relations $R_{<}$ and $R_{\leq}$ are transitive, but for example the relation:

$$R_{ab}^1 = \left\{ (a, b) : a, b \in \mathbb{N} \text{ and } a = b - 1 \right\}$$

is not, e.g., $(3, 4) \in R_{ab}^1$ and $(4, 5) \in R_{ab}^1$ but $(3, 5) \notin R_{ab}^1$.

Summary

- ▶ A **binary relation** between A and B is a subset of $A \times B$
- ▶ Examples: $R_<$, $R_\leq$ on the natural numbers, $R_\subseteq$ on $2^{\mathbb{N}}$
- ▶ **Reflexive**: $(a, a) \in R$ for every a
- ▶ **Symmetric**: $(a, b) \in R \Rightarrow (b, a) \in R$
- ▶ **Transitive**: $(a, b), (b, c) \in R \Rightarrow (a, c) \in R$

Partial order relations



Introduction: Partial order relations

- ▶ In this section we introduce the **antisymmetric** property and **partial order relations**
- ▶ We will see that “ $\leq$ ” and “ $\subseteq$ ” are partial order relations
- ▶ We will represent relations with **directed graphs** and discuss maximal elements
- ▶ Orderings are the basis for comparing numbers, sets and objects

Antisymmetric property

- ▶ A relation $R \subseteq A \times A$ is **antisymmetric** if:

$$\forall a, b \in A, \quad (a, b) \in R \quad \text{and} \quad (b, a) \in R \quad \Rightarrow \quad a = b$$

Example

- ▶ The relation $R_{\leq}$ is antisymmetric, since $a \leq b$ and $b \leq a$ imply $a = b$.

Definition 2: of partial order relation

*A reflexive, antisymmetric and transitive relation is a **partial order relation**.*

Example

- ▶ The relation $R_{\leq}$ is a partial order relation, but the relation $R_{<}$ is not, since it is not reflexive.

The relation $R_{\subseteq}$

Observation 1

The relation $R_{\subseteq}$ is a partial order relation

Proof (1/2)

We must prove that the relation is reflexive, antisymmetric and transitive.

- ▶ For it to be reflexive we must prove that $(S, S) \in R_{\subseteq}$, which is true since $S \subseteq S$.
- ▶ For it to be antisymmetric we must prove that if $S_1 \neq S_2$ then $S_1 \not\subseteq S_2$ or $S_2 \not\subseteq S_1$ or both, the negation of the desired property. Since $S_1 \neq S_2$:
 - either there exists some element that is in S_1 but not in S_2 , hence $S_1 \not\subseteq S_2$
 - or there exists some element that is in S_2 but not in S_1 , hence $S_2 \not\subseteq S_1$
 - or both of the previous options hold

Proof (2/2)

- For it to be transitive we must prove that $(S_1, S_2) \in R_{\subseteq}$ and $(S_2, S_3) \in R_{\subseteq}$ imply $(S_1, S_3) \in R_{\subseteq}$. Clearly, since $S_1 \subseteq S_2$ and $S_2 \subseteq S_3$, we have $S_1 \subseteq S_3$.



Partially ordered set

Definition 3: of partially ordered set

*A **partially ordered set** is a pair consisting of a set and a partial order relation defined on it.*

Example

- ▶ The sets of natural, rational or real numbers with the relation $R_{\leq}$ are partially ordered sets.
- ▶ The relation “is a descendant of” defined on a subset of people is a partial order relation (if we consider individuals as descendants of themselves).
- ▶ Consequently, the subset of people considered with the relation “is a descendant of” is a partially ordered set.

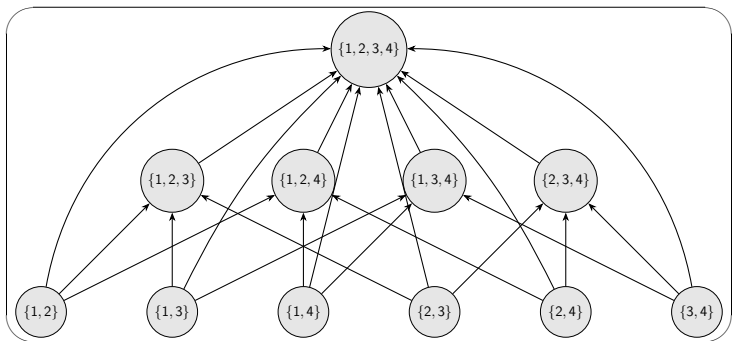
Representation with directed graphs

Relations can be represented by a **directed graph**, where the **vertices** are the elements of the set on which the relation R is defined and an **arc** (a, b) means that $(a, b) \in R$.

If the graph is acyclic, then the relation is a partial order relation.

Example

The directed acyclic graph associated with the relation $R_{\subseteq}$ on the set $\{1, 2, 3, 4\} \subseteq \mathbb{N}$ is the following (the sets consisting of a single number and the empty set are not shown in the figure):



Maximal elements

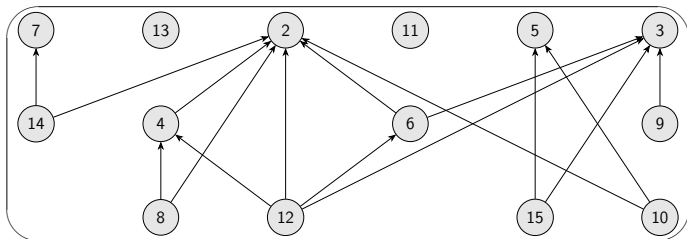
- ▶ In a partially ordered set there may be no single **maximum element**, that is, an element a such that:

$$\forall b \in A, \quad (b, a) \in R$$

- ▶ A partially ordered set may therefore contain several maximal elements a such that, for no $b \in A$ with $b \neq a$, we have $(a, b) \in R$.

Example

The directed acyclic graph associated with the partial order relation “is a divisor of” on the set $\{2, 3, \dots, 15\} \subseteq \mathbb{N}$ is the following:



Example

Given a set of boxes of different sizes, the relation “one box is contained in the other” on the set of boxes may have several maximal boxes, that is, boxes that are not contained in any other box.

Summary

- ▶ **Antisymmetric:** $(a, b), (b, a) \in R \Rightarrow a = b$
- ▶ **Partial order:** reflexive, antisymmetric and transitive (e.g. $R_{\leq}, R_{\subseteq}$; not $R_{<}$)
- ▶ **Partially ordered set:** a set with a partial order relation
- ▶ Relations are represented by **directed graphs**; acyclic graph $\Rightarrow$ partial order
- ▶ There may be several **maximal** elements and no maximum element

Total order relations

Introduction: Total order relations

- ▶ In this section we require that **every pair** of elements be comparable
- ▶ We will define total relations, total order relations and totally ordered sets
- ▶ The natural, rational and real numbers with “ $\leq$ ” are the fundamental example

Total relation

Definition 4: of total relation

A relation R on a set A is a **total relation** if:

$$\forall a, b \in A, \quad (a, b) \in R \quad \text{or} \quad (b, a) \in R \quad (\text{or both})$$

Example

- ▶ The relation $R_{\leq}$ is a total relation.
- ▶ The relation $R_{\subseteq}$ is not a total relation since, taking for example $S_1 = \{1, 2\}$ and $S_2 = \{2, 3\}$, $(S_1, S_2) \notin R_{\subseteq}$ and $(S_2, S_1) \notin R_{\subseteq}$.
- ▶ The relation “is a descendant of” is not a total relation since there exist pairs of individuals (a, b) for which neither a descends from b nor b descends from a .

Total order relation

Definition 5: of total order relation

*A partial order relation that is also a total relation is a **total order relation**.*

Example

- ▶ The relation $R_{\leq}$ is a total order relation.

Totally ordered set

Definition 6: of totally ordered set

*A **totally ordered set** is a pair consisting of a set and a total order relation defined on it.*

Example

- ▶ The sets of natural, rational or real numbers with the relation $R_{\leq}$ are totally ordered sets.

Summary

- ▶ **Total relation:** for every a, b we have $(a, b) \in R$ or $(b, a) \in R$
- ▶ **Total order:** a partial order that is also total
- ▶ $R_{\leq}$ is a total order; $R_{\subseteq}$ and “is a descendant of” are not total
- ▶ $\mathbb{N}, \mathbb{Q}, \mathbb{R}$ with $\leq$ are **totally ordered sets**

Functions

Introduction: Functions

- ▶ In this section we see **functions** as particular binary relations
- ▶ We will define domain, codomain, argument and value
- ▶ Functions are the central object of the whole Mathematical Analysis course



Definition 7: of function

Given two sets A and B , a binary relation f on A and B is a **function** if, for each $a \in A$, there exists one and only one $b \in B$ such that $(a, b) \in f$.

Definition 8: of domain and codomain

The set A is called the **domain** of f , and the set B is called the **codomain** of f .

Notation

- We write:

$$f : A \rightarrow B$$

and if $(a, b) \in f$, we write:

$$b = f(a)$$

since b is uniquely determined by the choice of a .

- Intuitively, the function f assigns an element of B to each element of A . No element of A is associated with two different elements of B . The same element of B can however be assigned to different elements of A .

Example

- The binary relation:

$$f = \left\{ (a, b) : a, b \in \mathbb{N} \text{ and } b = a \pmod{2} \right\}$$

is a function $f : \mathbb{N} \rightarrow \{0, 1\}$ since for all natural numbers a , there is exactly one value $b \in \{0, 1\}$ such that $b = a \pmod{2}$. For example,

$$0 = f(0), \quad 1 = f(1), \quad 0 = f(2), \dots$$

Example

- The binary relation

$$g = \left\{ (a, b) : a, b \in \mathbb{N} \text{ and } a + b \text{ is even} \right\}$$

is not a function, since for example $(1, 3)$ and $(1, 5)$ are both in g . In other words, for $a = 1$, we do not have one and only one b such that $(a, b) \in g$.

Argument and value

Definition 9: of argument and value

Given a function $f : A \rightarrow B$, if $b = f(a)$, we say that $a \in A$ is the argument of f and that $b \in B$ is the value of f associated with a .

- We can **define a function** by *directly defining its value* for all the *elements* of its *domain*.

Example

For example, we can define $f(n) = 2n$ for $n \in \mathbb{N}$, which means:

$$f = \{(n, 2n) : n \in \mathbb{N}\} \quad \text{and} \quad f : \mathbb{N} \rightarrow \mathbb{N}$$

Summary

- ▶ A **function** $f : A \rightarrow B$ is a binary relation in which every $a \in A$ is related to one and only one $b \in B$
- ▶ A is the **domain**, B the **codomain**; we write $b = f(a)$
- ▶ a is the **argument**, b the **value** of f at a
- ▶ $b = a \bmod 2$ is a function; “ $a + b$ even” is not
- ▶ A function can be defined by assigning the value to each element of the domain

Summary (1/2)

Binary relations:

- ▶ Subsets of a Cartesian product $A \times B$
- ▶ Properties: reflexive, symmetric, transitive, antisymmetric

Order relations:

- ▶ Partial order: reflexive, antisymmetric, transitive (e.g. $\leq$, $\subseteq$)
- ▶ Total order: a partial order in which every pair is comparable (e.g. $\leq$)
- ▶ Representation with directed acyclic graphs; maximal elements

Summary (2/2)

Functions:

- ▶ Relations that associate with every element of the domain one and only one element of the codomain
- ▶ Notation $f : A \rightarrow B$, $b = f(a)$
- ▶ Argument and value; definition of a function through its values